

WHAT IMPACTS
CALORIE
EXPENDITURE
DURING EXERCISE?

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1 INTRODUCTION

The fitness industry has become more dependent on data to optimise training programs for clients and gym members. Understanding the factors that drive the usage of calories during exercise is essential for designing effective individual and personal workout strategies (Wan, 2024). Despite a huge widespread assumption that specific workout types and higher intensity workouts lead to a greater calorie burn, research suggests that the relationship between variables in exercise and intensity might be more complex than we think.

This project aims to simulate a real-world business in which a fitness organisation is trying to find out and better understand which factors contribute to calorie burn and workout efficiency among its members. The problem area that will be attempted to clarify in this project is to identify the importance of workout durations, intensity (heart rate, etc.), workout types, and individual characteristics (gender, weight, etc.) in the determination of calorie expenditure.

1.1 Domain Background

Calorie expenditure from exercising is influenced by a combination of different factors. These could be factors such as exercise duration, the intensity of the exercise, body composition and metabolic responses. Traditionally, exercises with high intensity, such as interval training, have been associated with a higher calorie expenditure than other workout types. However, scientific evidence challenges this assumption.

A study from Yildiz & Akyildiz (2017) investigated the effects of steady state workout and interval running on acute calorie expenditure. Their findings revealed that there was no statistically significant difference between the two workout types in calorie expenditure when the duration of the workouts was held constant. An interesting finding in this study was that the steady state workout resulted in a higher heart rate and more consumption of oxygen and yet produced similar calorie expenditure. This challenges the common assumption that heart rate is a reliable predictor of calorie burn, and that it could be more advanced than that.

In addition to workout structure, individual characteristics, especially gender, play a key role when it comes to exercise outcomes. A study from Stojmanovska & Gontarev (2018) researched how gender differences in body composition change following a structured workout program. The results showed that there were significant reductions in body mass and fat percentage in both males and females. However, the male participants showed earlier and more pronounced reductions in body fat than the female participants, indicating that males have a faster physiological response to the workouts.

These differences are mostly explained by the variations in the body composition and hormonal regulation. Males typically have a higher amount of muscle mass and a higher energy expenditure, which allows males to burn fat faster (Del-Cuerpo et al., 2023). In contrast, females usually have a higher amount of fat percentage and a slower lipolytic response (the body's metabolic response for breaking down stored triglycerides in adipose tissue into free fatty acids and glycerol), especially in certain body parts. As a result of this, both genders benefit from the exercise, but the rates and mechanisms of adaptations may differ (Stojmanovska & Gontarev, 2018; Blaak, 2001).

Together, both studies are highlighting two important insights: firstly, that workout type alone may not determine calorie expenditure, and secondly, that individual characteristics, such as gender, will significantly influence both the effectiveness and the timing of the exercise outcomes.

2 PROJECT OBJECTIVES AND HYPOTHESES

The primary goal of this project is to identify key factors that influence calorie expenditure and workout efficiency among gym members by analysing workout data. This study aims to simulate a real-world business case for providing useful insights that can support data-driven decision-making within fitness organisations or personal trainers. More specifically, the objectives for this study are:

- To research the relationship between the workout duration and the total calorie expenditure
- To examine the impact of the average heart rate on calorie expenditure
- To compare different workout types and their efficiency in calorie expenditure
- To investigate how individual characteristics, such as gender, body weight and experience, influence the workout outcomes.

Based on the theoretical background and previous research mentioned in the domain background, the hypotheses are structured around three main categories: physiological responses, workout strategies, and individual differences.

When it comes to workout characteristics, it is expected that longer workout durations will lead to higher calorie expenditure in total (H1), while in terms of the workout types, it's not expected to influence the calorie expenditure significantly, when the duration of the workouts is controlled (H2). However, the type of workout may affect the calorie efficiency (H3).

Moving on to the physiological responses, it is hypothesised that the average heart rate will have a positive association with calorie expenditure, but this relationship may weaken when the workout duration is controlled (H4).

For the individual characteristics, body weight, gender, and experience level are expected to have a significant impact on the calorie expenditure from the individual participants (H5).

3 DATA DESCRIPTION

The dataset that was used in this study was collected from Kaggle and presents fictional workout tracking data that is designed to simulate real-world fitness data. The dataset is formatted as a CSV file and contains 972 observations from individual workout sessions from different gym members. Although the dataset aims to reflect realistic relationships between the workout variables and the calorie expenditure, it is important to note that the data is simulated instead of collected through real-life observations. As a result, some relationships between the variables may be more consistent or stronger than they normally would be in a real-life observation. The dataset includes variables that range from exercise data to individual characteristics. Some of the variables from this dataset will be more important than others, and some key variables from this dataset include:

- **Session Duration (Hours):** The workout session's total duration
- **Calories Burned:** The total expenditure of calories during a workout session
- **Workout Type:** Which type of exercise was performed in the workout session (strength training, cardio, yoga, HIIT, etc.)
- **Gender:** Which type of gender the participant is
- **Average Heart Rate (BPM):** The average heart rate that is recorded during the workout session
- **Weight (kg):** The body weight of the participants
- **BMI:** Body Mass Index, which represents the participant's body composition
- **Experience Level:** The participants' workout experience level

Since the workout durations from each participant are different, a new variable called **Calories_Per_Hour** was created to account for the differences in the workout durations. This variable was created by dividing the total calorie expenditure during a workout session by the duration. The main reason for making this variable was to allow for fair comparisons between sessions of different lengths. By standardising the calorie expenditure for time, instead of the whole workout duration, it becomes possible to see the effect

of other variables like heart rate and workout type. This is because the workout duration could have a significant impact on the heart rate and calorie expenditure from different workout types. By expressing calorie expenditure on a per-hour basis, it becomes possible to make more meaningful comparisons between workout sessions of different lengths.

4 EXPLORATORY DATA ANALYSIS (EDA)

4.1 Relationship Between Workout Duration and Calorie Expenditure

The first analysis investigates whether there is a relationship between the duration of the workout and total calorie expenditure. A scatterplot was created with session duration as the independent variable and calories burned as the dependent variable.

The scatterplot (see figure 1) reveals a strong and positive relationship between the workout durations and the calorie expenditure. As the work duration increases, the number of calories also increases significantly. The trendline shows that the datapoints clearly follow an upward trend, which indicates that workout duration is a strong predictor for calorie burn.

In addition to this, the overlap between the workout types suggests that the workout duration may have a bigger influence on total calorie expenditure than the specific type of exercise performed. While there is some variability at the longer durations, the overall relationship is still very consistent throughout the dataset. The findings from this analysis support hypothesis 1 (H1), which suggests that longer durations of workouts will have a positive association with higher calorie expenditure.

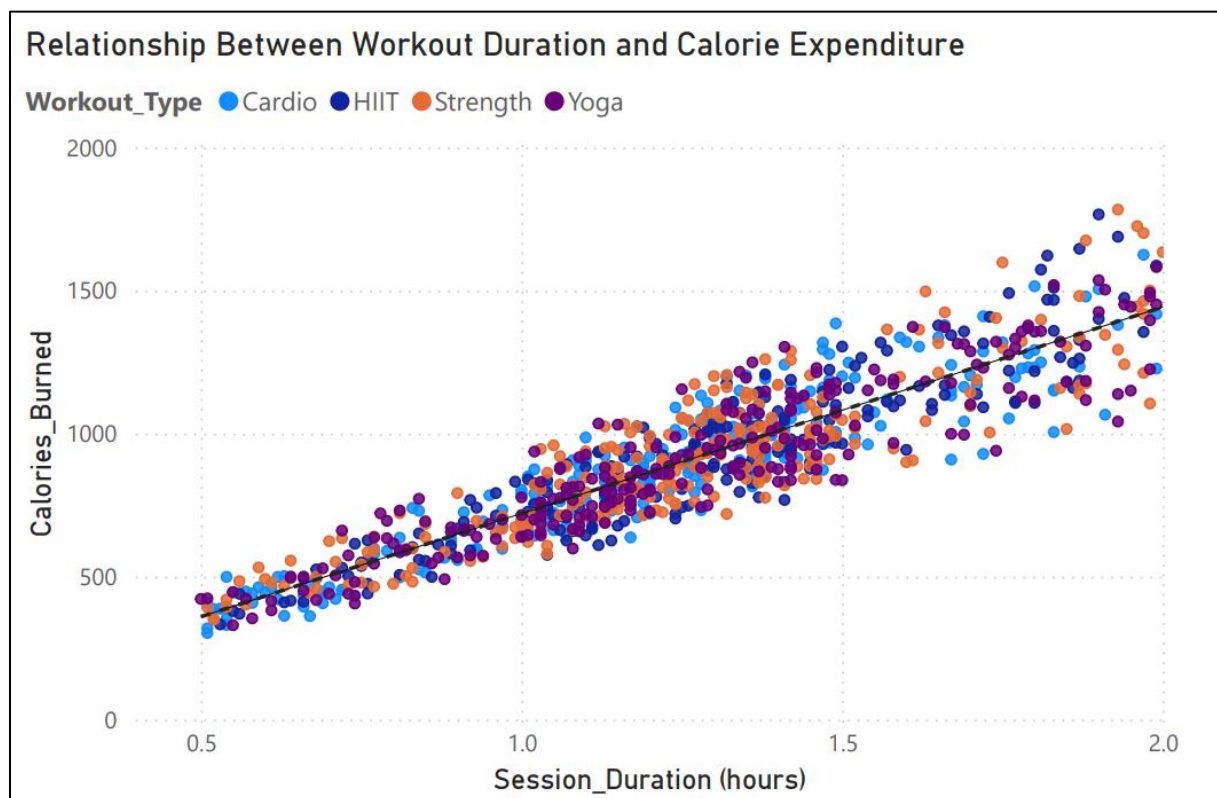


Figure 1. Scatterplot illustrating the relationship between workout duration (hours) and total calories burned.

4.2 Relationship Between Average Heart Rate and Calorie Expenditure

The second analysis that was performed tries to investigate the relationship between the average heart rate and the total calorie expenditure. A scatterplot was created with average heart rate (BPM) as the independent variable and calories burned as the dependent variable.

This visualisation indicates that there is a positive relationship between the average heart rate and calorie expenditure, where higher heart rates generally correspond to increased calorie expenditure. However, if we compare it to the relationship that was observed between the workout duration and calorie expenditure in the previous analysis, this relationship appears to be a lot weaker and much more variable.

A large spread of data points can be observed on the graph (see figure 2), indicating that individuals with similar heart rates still show significant differences in their expenditure of calories. This suggests that while

heart rate may contribute to calorie expenditure, factors such as workout duration, body weight, and workout type may also have a significant role.

The overlap between the workout types indicates that the heart rate alone may not be adequate to explain the differences in the calorie expenditure across the workout categories.

The findings from this analysis support hypothesis 2 (H2) partially, which suggests that average heart rate is positively associated with calorie expenditure. However, the weaker relationship suggests that the effect of the average heart rate may be less influential, especially when compared to other factors, such as the workout duration.

Effect of Heart Rate on Calorie Expenditure

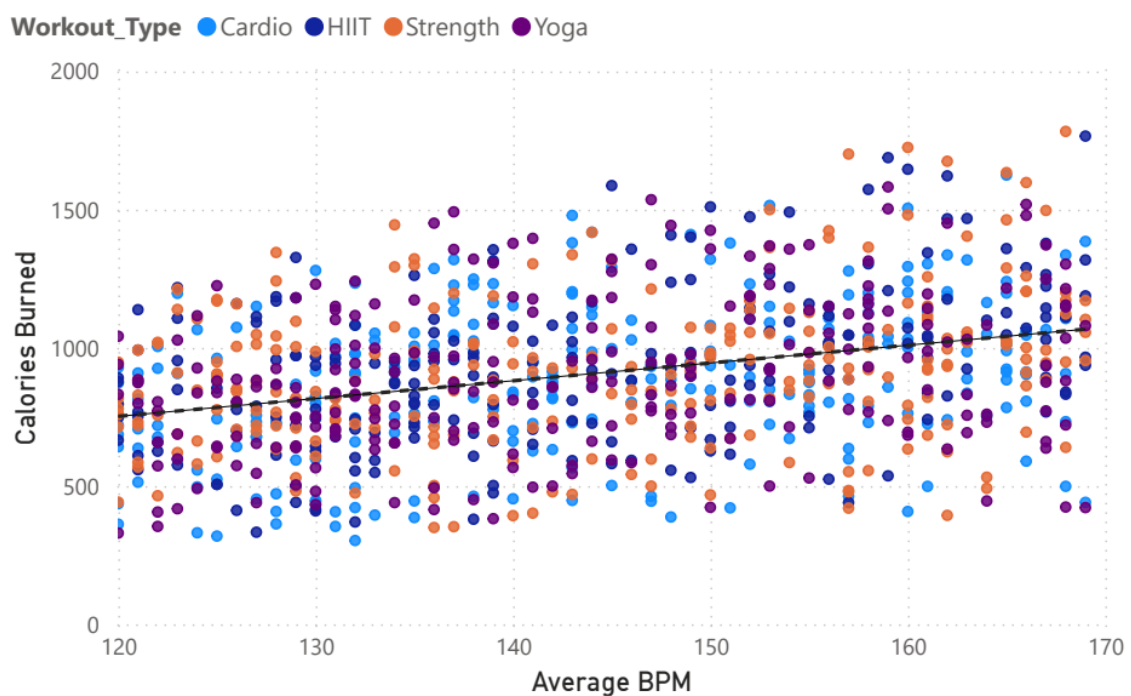


Figure 2. Scatterplot illustrating the relationship between the average heart rate (BPM) and the total calories burned across workout sessions.

4.3 Comparison of Workout Efficiency Across Different Workout Types

In the third analysis, the differences in workout efficiency were examined across the exercise categories. To account for the variations of the workout durations from the participants, the newly created variable “Calories_Per_Hour” was used instead of the original variable “Calories_Burned”. This allowed for a more precise and meaningful comparison between the workout types.

For this analysis, a clustered column chart was created to compare the average calorie expenditure per hour across different workout types (see figure 3). Figure 3 presents the average calorie expenditure per hour for each of the workout types. These results indicated that the workout type with the most calorie expenditure was **Strength training** (258kcal/hour), followed closely by **Cardio** with the highest expenditure rate (255 kcal/hour). **Yoga** had a lower average calorie expenditure (239 kcal/hour), while **HIIT** had the lowest recorded average calorie expenditure rate (221 kcal/hour)

The findings from this analysis suggest that workout efficiency may differ between the exercise categories. HIIT is often perceived as the most intensive form of workouts, and interestingly, it came out with the lowest expenditure rate. Instead, Strength training and cardio were the workout types that came out with the highest calorie expenditure rate.

The differences that were observed between the workout types support Hypothesis 5 (H5), which suggested that the workout type influences calorie efficiency. However, the relatively small differences between some of the workout categories suggest that the workout type alone may not be a dominating factor for the calorie expenditure rate. Other factors like workout duration, heart rate, body composition and other individual characteristics may have contributed to the results.

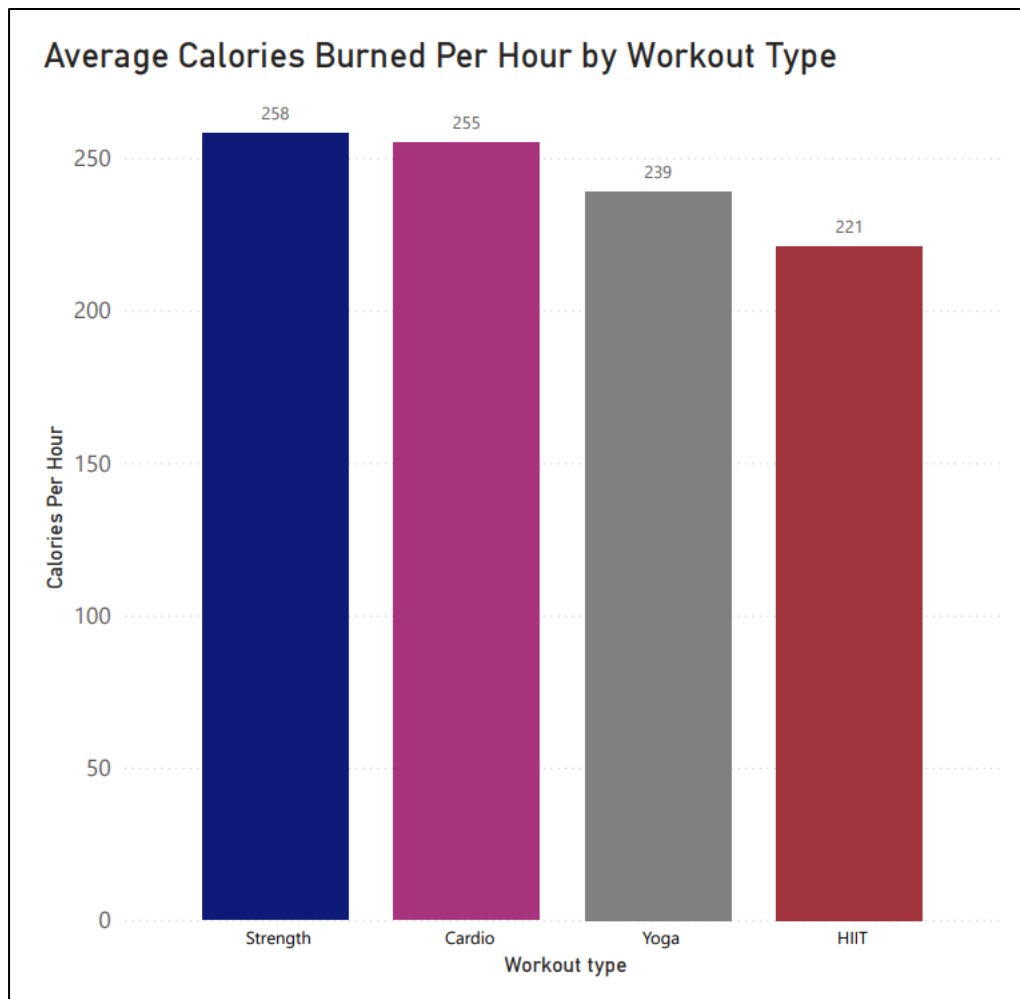


Figure 3. Clustered column chart illustrating average calories burned per hour across different workout types.

4.4 Gender Differences in Calorie Expenditure

The fourth exploratory analysis investigates whether there is a difference in calorie expenditure between male and female participants. The variable “Calories_Per_Hour” was also used in this analysis to account for the differences in the workout durations. For this analysis, a clustered column chart was created to easily visualise the differences between the average calorie expenditure for male and female participants (see Figure 4).

The results indicate that the male participants had an average calorie expenditure of 511 calories per hour, while the female participants had an average of 462 calories per hour. This suggests that the male participants had a higher expenditure of calories per hour in this dataset.

The findings for this analysis support parts of Hypothesis 3 (H3), which suggested that individual characteristics, including gender type, influence workout outcomes and calorie expenditure.

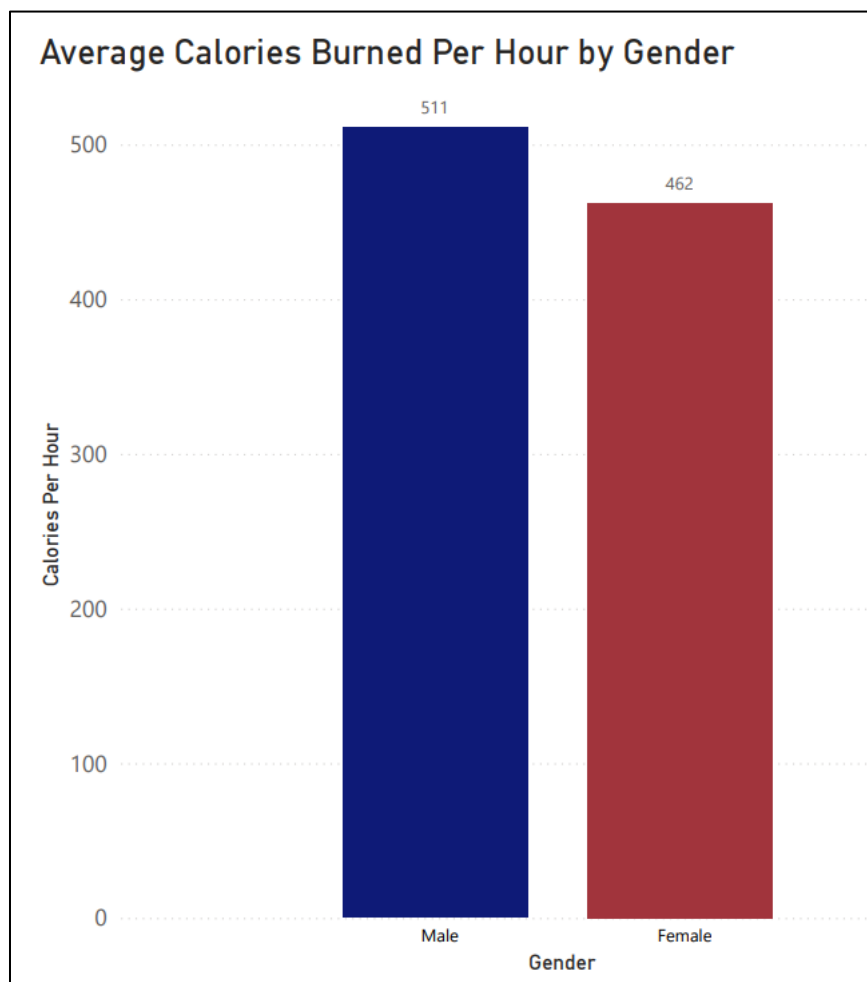


Figure 4. Clustered column chart illustrating difference in average calories burned per hour for males and females.

4.5 Relationship Between Body Weight and Calories Burned

The fifth analysis aims to investigate whether there is a relationship between body weight and total calorie expenditure. A scatterplot was chosen to visualise this relationship, using body weight as the independent variable and calories burned as the dependent variable.

The visualisation is showing a weak but positive relationship between body weight and calorie expenditure. As the body weight increases, there is a slightly increasing trend in calorie expenditure, which can also be seen by the trendline. However, there is a lot of variation in the observations throughout the dataset.

The huge spread of observations suggests that body weight alone might not be a strong predictor of calorie expenditure. The chart shows that individuals with similar body weights often demonstrate huge differences in the calorie expenditures, which indicates that other factors like the workout duration, heart rate, and workout type may be in play, and have a greater impact on the calorie expenditure rates.

The inclusion of gender highlighting in the chart clearly shows the distribution of body weight between male and female participants. The male participants clearly occupy the higher weight ranges in the dataset, which could be a possible contributing factor to why the male participants demonstrated higher average calorie expenditure in the previous analysis, seen in Figure 4. These findings provide additional support for Hypothesis 3 (H3).

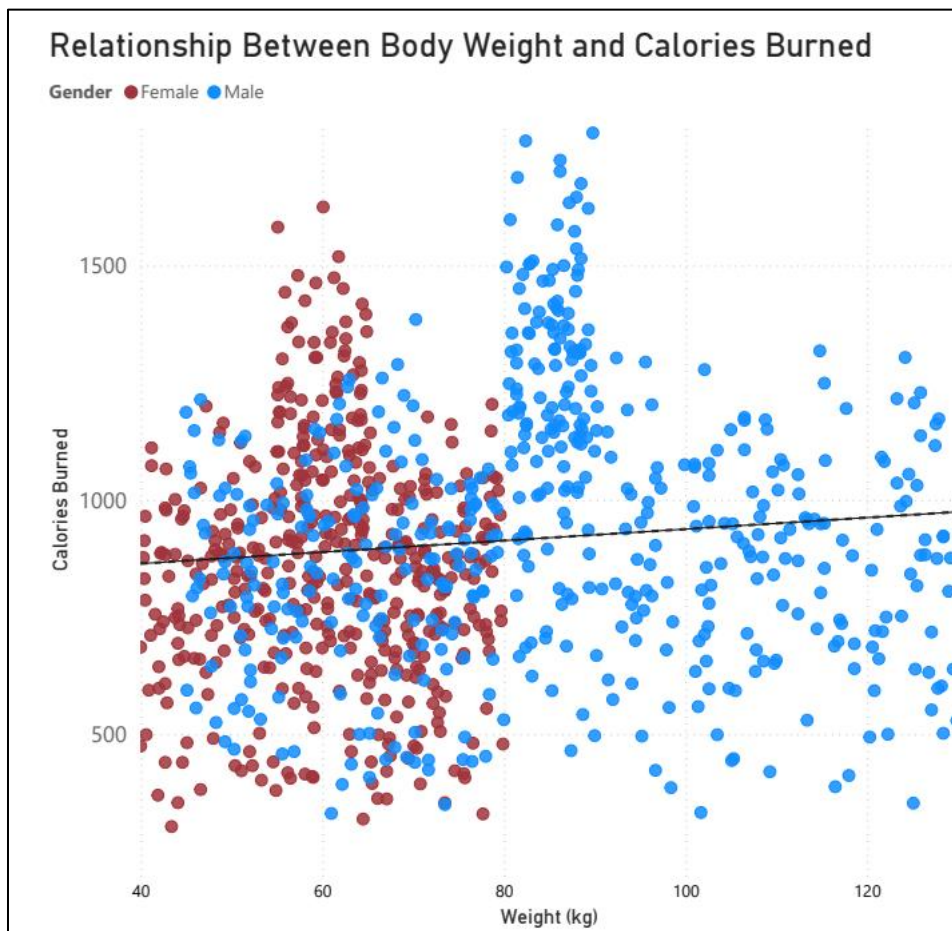


Figure 5. Scatterplot illustrating the relationship between body weight (kg) and total calories burned, with males and females contrasted.

4.6 Relationship Between Experience Level and Calorie Expenditure

The sixth and last exploratory analysis looks into experience level and investigates if there is a relationship between the participants' experience and calorie expenditure. For this analysis, another clustered column chart was created to visualise the average calorie expenditure differences between the three experience levels in the dataset (see Figure 6).

The results revealed that there was a clear relationship between the participants' experience level and their calorie expenditure rates. Participants with the experience level of 1 burned an average of 730 calories per workout session, while participants with the experience level of 2 burned an average of 900 calories per

session. The highest calorie expenditure rate was observed among the participants with the experience level of 3, who burned an average of 1260 calories per session.

The findings from this analysis suggest that more experienced individuals tend to expend more calories in their workouts. A possible explanation for this could be that experienced individuals may be able to train with higher intensities, perform more advanced and demanding exercises, or sustain physical activity for a longer period of time during the workouts. As a result of this, they may expend more energy during their workout sessions compared to individuals with less experience.

However, it is important to note that this analysis was based on the variable “Calories_Burned”, which looks at the total calorie expenditure for the whole session, rather than the calorie expenditure per hour. Therefore, parts of the observed difference in the analysis might be explained by differences in workout durations across the experience levels. More experienced participants may have longer workout sessions, which could be a contributing factor to why they have a higher overall calorie expenditure rate. The results from this analysis further support Hypothesis 3 (H3).

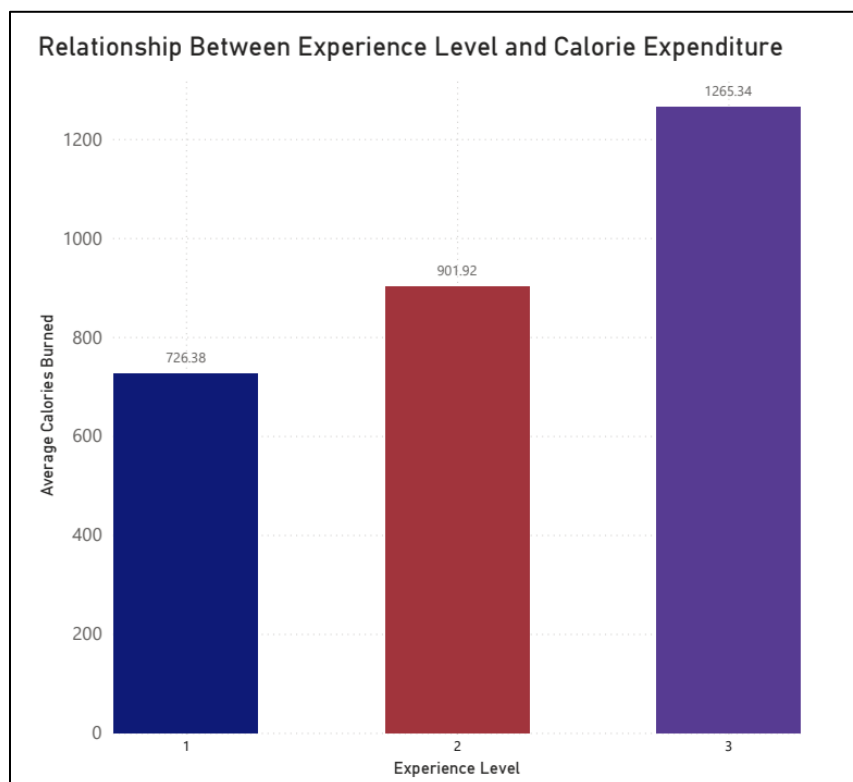


Figure 6. Clustered column chart illustrating the average calories burned across three different experience levels from 1 (lowest) to 3 (highest).

5 DATA PRE-PROCESSING

Before starting with the exploratory analysis, the dataset was prepared and reviewed to make sure that it was suitable for analysis. Data preprocessing is important to improve the data quality and support data visualisation.

5.1 Data Quality Assessment

Firstly, the dataset was inspected for any missing data, duplicate values or any other potential inconsistencies. This step was crucial to make sure that the analysis was based on complete and reliable observations.

There were no significant missing values identified in the dataset, and no duplicate workout sessions were found. Additionally, the variables were reviewed to make sure that numerical values in the variables, such as **Calories Burned**, **Heart Rate**, **Workout Duration**, etc., were stored using numerical values. Variables like **Gender** and **Workout Type** were also verified to text values, to make sure they would be correctly analysed in Power BI. Since there were no major data quality issues that were identified, the dataset only required limited data cleaning before the analysis could begin.

4.2 Feature Engineering

The variable **Calores_Per_Hour** was created to support the analysis and to draw out rightful insights. This variable was created by dividing the total calories burned during a workout session by the workout session duration (in hours).

The reason for making this variable was to account for the differences in the workout duration for each of the participants. Since their workout durations are different, the total duration of the workout will have a strong impact on the total expenditure from the sessions. Comparing total calories burned alone would not be a fair comparison between workouts that have different lengths. By expressing calorie expenditure relative to the duration, it became possible to compare the workout efficiency across other variables.

4.3 Technical Transformations

To assign a unique identifier to each observation in the dataset, an additional index value was created. The index value was used in Power BI to ensure that each workout session was presented individually in the scatterplot analyses. Without a unique index for each workout, there is a chance that Power BI could combine the observations when creating the scatterplots. Therefore, it was important to create the index column to improve the accuracy of the exploratory analyses and to allow the patterns and relationships between the variables to be observed effectively.

Overall, there were limited pre-processing steps that were required in this data set due to its clean structure. The creation of the variable “Calories_Per_Hour” and the index column provided important enhancements for the analysis.

5 TRENDS, PATTERNS AND ANOMALIES

The exploratory analysis found several trends and patterns that were notable regarding calorie expenditure and workout performance. Some of the findings aligned closely with the proposed hypotheses and the existing research, but others provided some unexpected insights into the factors that influenced the workout outcomes.

5.1 Workout Duration as The Strongest Predictor of Calorie Expenditure

The clearest pattern that was found through the analysis was the strong positive relationship between the workout duration and calorie expenditure. The scatterplot that was created visualised a clear and consistent upward trend, indicating that the participants who had longer workouts generally achieved a higher calorie consumption. Workout duration appeared to have the strongest relationship with the total calorie usage compared to all the other variables in the dataset. The findings from this analysis suggests that the workout duration is one of the more important factors that impacts the total calorie expenditure rate during a workout session.

5.2 The Influence of Heart Rate on Calorie Expenditure

A weaker, but positive relationship was also observed between the participants average heart rate and their calorie expenditure. The participants with higher average heart rates generally burned more calories during their workout sessions. This relationship was still noticeably weaker than the relationship between the workout duration and calorie expenditure. Some variations between the calorie expenditure rates from the participants with similar heart rates were observed, which suggests that heart rate alone is not sufficient to explain the differences in the calorie expenditure. The findings from this analysis align with the study from Yildiz & Akyildiz (2017), which indicates that heart rate does not always have an impact on greater calorie consumption.

5.3 Differences in Workout Efficiency Between Exercise Types

The analysis of workout efficiency from the different workout types revealed some measurable differences. Strength training and cardio were the two workout types that showed the highest rates of calorie expenditure, while yoga showed some slightly lower values. Surprisingly, HIIT was the workout type that showed the lowest average calorie expenditure within the dataset. The findings from this analysis were unexpected, as high-intensity training is often associated with higher calorie-burning potential. The results from this analysis suggest that workout type may have an impact on the efficiency of calorie expenditure, but it appears to be smaller than is commonly assumed.

5.4 Gender Differences in Calorie Expenditure

The male participants had a higher average calorie consumption per hour than the female participants. This aligns with previous studies that males generally possess a higher amount of muscle mass, which contributes to a greater calorie expenditure (Del-Cuerpo et al., 2023). Although the observed outcomes were moderate, the results still support the view that gender can influence the workout outcomes and calorie expenditure.

5.5 The Influence of Individual Characteristics

Through the analysis of the individual characteristics, the experience level of the participants showed one of the strongest relationships with the calorie expenditure. The participants with greater experience levels showed consistently greater average calorie expenditure than the less experienced participants.

Body weight also showed a positive relationship with calorie expenditure, but this relationship was weak. Participants with the same body weights showed significant variation in calorie expenditure, which indicates that other factors would more strongly impact the expenditure rates.

5.6 Anomalies and Unexpected Findings

Some unexpected observations were found during the analysis. The most notable finding was that HIIT did not come out as one of the highest calorie efficiency workouts, despite it being commonly perceived as one of the most demanding workout types. Additionally, the relationship between body weight and calorie expenditure was also weaker than expected.

There were no major outliers or any other abnormal observations identified in the dataset. The overall distribution of the observations appeared to be consistent. Since this dataset is simulated and not collected from a real-world scenario, the dataset may be more consistent and have less variability than a typical dataset observed in a real-world situation.

Overall, the findings suggest that the workout duration, experience level and gender are the most influential factors when it comes to calorie expenditure in this dataset. Workout type and heart rate appear to have more moderate impacts on calorie consumption than commonly assumed.

6 DISCUSSION AND REAL WORLD APPLICATIONS

The primary goal for this task was to identify factors that influence the calorie expenditure, and workout efficiency for gym members through various analysis. The findings from the analysis suggest that there are multiple variables that can contribute to workout outcomes, but that some factors have a much greater impact than others.

The strongest relationship that was observed in the analysis was between workout duration and calorie expenditure. The participants who exercised for longer periods of time consumed the most calories regardless of the workout type. This suggests that fitness and sport organisations should account for workout duration when designing exercise programs for members and clients, when weight management is the primary goal.

The analysis also showed a positive relationship between the heart rate and the calorie consumption. This relationship was a lot weaker but still remains as a useful indicator for calorie expenditure, though it should not be used as a solo measure for workout effectiveness. Therefore, fitness professionals may benefit from considering multiple performance-related factors under the evaluation of workout effectiveness and outcomes.

Differences were also observed when it comes to the workout types. Interestingly enough, HIIT did not demonstrate the highest calorie consumption per hour of the different workout types within this dataset. Instead, strength training and cardio were the workout types which showed the greatest calorie efficiency. The results from this show that highly intensive workouts may not always be necessary to accomplish optimal workout outcomes.

The analysis demonstrated further that individual characteristics had a significant influence on the workout outcomes. Male participants had a higher average calorie consumption per hour than the female participants, which supports previous studies highlighting differences in gender body composition and energy usage. In addition to this, participants who had a higher experience level consistently showed greater calorie expenditure than the less experienced participants. These findings may suggest that training experience contributes to improved workout performance through increased intensity, greater capacity or better techniques.

The findings from this analysis can have several practical applications. Fitness and sports organisations can use the information gathered from the analysis as inspiration to develop more personalised training programs for members and clients. Recommendations can be based on individual characteristics such as gender, body weight or experience level, rather than giving out a standard training program to all members. By tailoring the workout program to each individual's physical profile and goals, chances for better workout outcomes for the members and clients will also increase.

Further, the findings may help fitness organisations in a business-related manner, by improving the members' engagement and reducing churn rate. By knowing which factors will contribute the most to achieving the members' goals, fitness trainers can provide more realistic expectations and guidance based on evidence. By helping members achieve their goals more effectively, it can increase motivation and long-term participation in the fitness programs.

Lastly, it's important to acknowledge that the dataset that was used in this analysis is simulated, rather than collected from real-world tracking devices. The dataset reflects somewhat realistic fitness-related relationships, but there is a lot of data in this dataset that doesn't fully support other data gathered in a real-world situation. Therefore, the findings from this study may not fully represent realistic relationships in workout outcomes and the complexity and variability of actual human physical responses.

7 CONCLUSION

The aim of this project was to investigate different factors that influence the calorie output and workout efficiency among gym members. By analysing a simulated fitness dataset, the project looked at the effects of workout duration, heart rate, workout type, gender, body weight and the participants' workout experience level on workout outcomes.

The findings from this project suggest that the workout outcomes are influenced by a combination of factors, rather than a single variable. This highlights the importance of adopting data-driven approaches when designing workout programs and strategies. The findings may help fitness and sport organisations to develop more personalised recommendations for their members and clients.

This project has given valuable experience in data preparation, data visualisation, and analytical thinking. Even though the dataset was simulated, the project has given meaningful insights into how powerful fitness-related data can be to gain knowledge of how individual humans respond to different workout strategies, and demonstrated the practical value of data analysis in a real-world context.

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